

# **The Impact of Communicating Multiple Goals: Evidence from a Field Experiment in an Energy Corporation**

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## **Abstract**

Firms often pursue and communicate multiple goals, although theory highlights the challenges of doing so. I study how communicating multiple goals affects employee performance, and argue that goal communication changes what employees attend to and allocate effort on. I design a field experiment across frontline employees in a multinational energy corporation as they are evaluated on a key procedure, randomly varying (1) whether they are communicated a single goal of safety or multiple goals of safety and efficiency, and (2) whether they are additionally provided with an observation of high-performing peers. Communicating multiple goals Pareto-improves performance, and these gains are largest when employees have room to improve on both dimensions. However, when the same multiple-goal communication is paired with observation, the performance gains from either intervention weaken. The results suggest that goal communication can improve performance when it directs attention and effort toward multiple dimensions inside the frontier, but can also complicate learning from observation by shaping what employees attend to when deciding what to copy from observation.

**Keywords:** multiple goals, incentives, best practices, field experiment, communication

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## 1. INTRODUCTION

Firms often pursue and communicate multiple goals to employees (e.g., Gubler, Larkin, & Pierce, 2016; Obloj & Sengul, 2020). Factories seek to increase output while keeping quality defects low, service businesses ask employees to increase customer satisfaction while remaining efficient, and many companies increasingly articulate social goals such as safety and environmental sustainability alongside financial ones (Burbano, 2016; Dimitriadis et al., 2017; Gubler, Larkin, & Pierce, 2018; Lee, Adbi, & Singh, 2020; Battilana et al., 2020).

Despite this empirical reality, much research warns against communicating multiple goals for employees to pursue, especially when goals are weakly or negatively correlated.<sup>1</sup> Economic theory on multitasking suggests that incentivizing multiple dimensions can distort effort when some dimensions are less measurable (Kerr, 1975; Holmstrom & Milgrom, 1991; Baker, 1992).<sup>2</sup> Organizational scholars similarly argue that multiple goals can create confusion and ambiguity for decision-makers, reducing performance even when dimensions are observable (Cyert & March, 1963; Porter, 1996; Jensen, 2002; Hu & Bettis, 2018; Levinthal & Rerup, 2020). These theories often imply that firms should prioritize a single goal, separate goals across units, or sequence attention to one goal at a time (Ethiraj & Levinthal, 2009; Stevens et al., 2015).

This contrast between theory and empirical reality raises a central question: how does communicating multiple goals affect employee performance? Empirical evidence remains limited, in part because goal communication inside firms is difficult to observe and measure, and also because firms that communicate multiple goals are likely to differ systematically from those that do not. These challenges make it difficult to know whether multiple-goal communication itself affects performance, and if so, when it helps or hurts.

I argue that its effects are unlikely to be uniform. First, they may depend on where employees are relative to a productivity frontier. When employees are near the frontier, improving one dimension may require sacrificing another. But when employees are inside the frontier, communicating multiple goals may improve performance by directing attention and effort toward dimensions that can both be improved. Second, performance effects may depend on the information available to identify how to improve. Goals are rarely communicated in isolation; employees often receive goals together with feedback, examples, templates, or observations of high performers. Understanding goal communication therefore requires examining not only which goals are communicated, but also how that communication shapes what employees take from the information available to them.

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<sup>1</sup> This applies to cases when multiple goals lack positive complementarities, such as a high volume of output and good quality. When goals are mutually reinforcing, then pursuing either one can increase the other.

<sup>2</sup> Holmstrom and Milgrom (1991) suggest that whenever tasks are separable, they can be carried out by different employees at different times. For example, if quality control can be allocated to a different worker from that in the production line that is maximizing output, the problem posed by incentives based on quantity can be mitigated or even eliminated.

I empirically evaluate these ideas through a field experiment among frontline employees in three subsidiaries of a Fortune 200 energy corporation. The experiment addresses the measurement and selection problems by explicitly varying which goals are communicated to employees, randomly assigning that communication across the organization, and cross-randomizing whether employees receive this communication alone or together with best practice observation of high-performers. This design allows me to examine whether multiple-goal communication improves performance when employees act directly on feedback, whether its effects are strongest when employees have room to improve safety and efficiency jointly, and whether the same communication weakens gains from best practice observation by making it harder to infer what to copy and why.

The experiment was implemented among all pairs of linemen in the three subsidiaries as they were retrained and evaluated on grounding, a key procedure that connects electrical lines to the earth to de-energize them before and after any work on live lines. Grounding was central to both safety and efficiency: it had to be performed correctly to prevent serious accidents, but it was also time-consuming and repeated frequently. As in many settings where safety and efficiency are viewed as competing demands (Bennett, 2014; Greve & Seidel, 2014; Gaba & Greve, 2019), the company viewed the safest version of the procedure as trading off with time at the productivity frontier.

The experiment cross-randomized two treatment arms. The first varied goal communication: linemen received either supervisor feedback emphasizing safety alone or both safety and efficiency. This arm identifies the effect of communicating multiple goals on performance and allows me to test whether the effect is larger among pairs with more room to improve. The second varied whether linemen also observed a best practice video of high-performers from a different subsidiary completing the same grounding step, to capture a common way in which companies implement best practice transfer to enable learning (Inkpen 2005, 2008, Argote and Fahrenkopf 2016). The video contained no instructional audio, no verbal discussion of safety or efficiency, and no labels identifying which observed actions improved which goal. This arm examines whether goal communication changes what employees take from the same observed behavior, and was cross-randomized with the first.

This design resulted in four experimental conditions, where treatment content was layered on top of the *Control* video that communicated safety through supervisor feedback: (1) *Multiple Goals* added explicit communication about efficiency; (2) *Best Practice* added unlabeled observation to the *Control* video; and (3) *Best Practice + Multiple Goals* added the same unlabeled observation to the *Multiple Goals* video, which itself consisted of the *Control* video plus efficiency communication. The design therefore creates two key comparisons. *Control* versus *Multiple Goals* tests whether communicating multiple goals changes performance when employees act directly on feedback. *Best Practice* versus *Best Practice + Multiple Goals* tests whether communicating multiple goals changes what employees take from the same observed behavior. All treatments were implemented digitally as videos, with all 270 linemen pairs randomly assigned to view one of the four types of videos. After watching their

assigned video, all linemen performed the grounding procedure again and were confidentially evaluated on both efficiency (time taken to complete) and safety (a score based on an internal company rubric).

I find that communicating multiple goals improved efficiency without reducing safety. Although safety and efficiency were negatively correlated at baseline, suggesting that safety and efficiency can trade off at the margin, linemen assigned to *Multiple Goals* reduced completion time by 11% relative to *Control*, with no economically or statistically significant decline in safety. These gains were largest among pairs that were slower and less safe at baseline, consistent with the interpretation that multiple-goal communication is most beneficial when employees are inside the productivity frontier and have room to improve on both dimensions.

However, these gains disappeared when multiple-goal communication was combined with best practice observation. *Multiple Goals* and *Best Practice* each improved performance when provided alone, but pairs assigned to *Best Practice* + *Multiple Goals* did not benefit from the same observation video. This pattern suggests that goal communication shapes not only what employees try to improve, but also what they take from observed behavior. When observation is unlabeled, making multiple goals salient may increase ambiguity about which actions to copy, which goals those actions advance, and whether those actions will translate to one's own performance.

These findings contribute to the growing literature on the challenges of managing multiple goals (Ethiraj & Levinthal, 2009; Hu & Bettis, 2018; Gaba & Greve, 2019; Obloj & Sengul, 2020). The results show that multiple-goal communication is not uniformly harmful or beneficial. Its performance consequences depend on whether employees have room to improve on multiple dimensions and on the information available to them to identify how to improve. Multiple-goal communication can improve performance relative to single-goal communication when employees act on direct feedback inside the productivity frontier, but it can weaken gains from observation when employees must infer what to copy from unlabeled examples.

This paper also contributes to the literature on best practice transfer by showing that the value of observing high performers depends on whether employees can infer what to copy and why. Best practice observation provides information about what high performers do, but not necessarily about which actions drive performance, which goals those actions advance, or what employees should copy. Prior research shows that information on best practices can improve adoption and performance (e.g., Bloom & Van Reenen, 2007; Bloom et al., 2013; Chatterji et al., 2019), while also highlighting the challenges of knowledge transfer arising from the individuals involved, the task, and the tools used for transfer (Argote & Miron-Spektor, 2011; Argote & Fahrenkopf, 2016). The findings from this paper suggest that the same observed behavior can lead to different performance gains depending on what employees are asked to attend to when interpreting it. As firms increasingly use data, examples, and observation-based information to learn from across the internal organization (Brynjolfsson &

McElheran, 2019) as well as the broader market (Kim, 2021), these findings suggest that more information may not always improve performance when employees face multiple goals. Without guidance about which observed actions advance which goals, best practice observation may become harder to translate into improved performance.

## **2. CONCEPTUAL MOTIVATION AND RELATED LITERATURE**

This section develops the paper's argument in three steps. I first review why existing theory generally predicts that communicating multiple goals should reduce performance. I then argue that this prediction depends on where employees are relative to a productivity frontier: when employees have room to improve on multiple dimensions, communicating multiple goals may direct attention and effort toward joint improvement. Finally, I argue that the effects of goal communication also depend on the information available to identify how to improve. When employees learn from observation, communicating multiple goals may make it harder to infer which observed actions matter, which goals they advance, and what to copy.

### **2.1 The impact of communicating multiple goals**

Theory across economics and strategy generally predicts that asking employees to pursue multiple weakly or negatively correlated goals will reduce performance. Agency theory highlights that multiple goals can distort effort when some dimensions are more measurable than others (Kerr, 1975; Holmstrom & Milgrom, 1991; Baker, 1992). In the canonical Holmstrom and Milgrom (1991) model, incentivizing multiple tasks can cause employees to reallocate effort toward more easily measured dimensions and away from less measurable ones. Research building on this logic suggests that improved measurement or better task and contract design can mitigate these distortions, for example by separating tasks or aggregating them depending on their relative measurability.

Organizational theory raises a related but broader concern: even when goals are observable, communicating multiple goals may reduce performance by increasing confusion, ambiguity, and complexity for decision-makers (Simon, 1955; Jensen, 2002; Ethiraj & Levinthal, 2009). When goals are not perfectly aligned, employees may struggle to determine which goal to prioritize, how to allocate attention across goals, and which actions advance one goal without undermining another. This literature therefore often implies that firms should choose a single goal, separate goals across organizational units, or focus on goals sequentially rather than asking employees to pursue several at once (Ethiraj & Levinthal, 2009; Joseph & Wilson, 2018).

Despite this rich theoretical literature, causal evidence on the performance effects of communicating multiple goals remains limited. One challenge is measurement: goal communication is difficult to observe inside organizations, leading empirical studies to infer the presence of multiple objectives from industry knowledge, survey responses, or other organizational choices (e.g., Hu &

Bettis, 2018; Gaba & Greve, 2019; McCann & Vroom, 2014). A second challenge is selection: firms that communicate or pursue multiple goals may differ systematically from those that do not. For example, firms may be more likely to emphasize multiple goals when they are underperforming, when they face more complex tasks, or when performance already depends on several dimensions. As a result, correlations between multiple goals and lower performance may not identify the causal effect of communicating multiple goals.

Existing causal evidence, largely from economics, provides a more mixed picture than standard multitasking theory might suggest. Rather than directly examining goal communication, this work typically studies whether incentives or performance pressure on one dimension lead workers to neglect another. Several studies find limited evidence of such effort distortion: workers across various contexts from the factory to educational and emergency response settings improve targeted outcomes without sacrificing other dimensions of performance (Al-Ubaydli et al., 2015; Hossain & Li, 2014; Englmaier et al., 2017; Chetty et al., 2014; Glewwe et al., 2010; Muralidharan & Sundararaman, 2013; Gubler, Liu, & Roman, 2021; Kim & Slonim, 2021).<sup>3</sup> Hong et al. (2018), for example, find that only a subset of workers trade off quality for quantity within a factory setting. This evidence raises the possibility that multiple objectives do not always hurt performance, but it does not directly test how communicating multiple goals affects employees' overall performance.

This paper builds on this research by directly testing how communicating multiple goals causally affects performance. It provides evidence from a randomized controlled trial within three companies inside a large corporation, where random assignment allows me to identify the effect of communicating multiple goals while holding broader organizational goals, tasks, and evaluation criteria constant.

## **2.2 Goal communication and the productivity frontier**

I argue that the effect of communicating multiple goals depends in part on where employees are relative to a productivity frontier. Much of the theoretical concern about multiple goals implicitly assumes that employees or firms are already operating on or near the frontier, where improving one dimension requires sacrificing another. Yet a large empirical literature documents persistent productivity dispersion across firms, plants, and units, even after accounting for hard inputs such as technology and capital intensity. In relatively homogeneous industries such as ready-mixed concrete, boxes, gasoline, ice, and sugar, Syverson (2004) and Foster, Haltiwanger, and Syverson (2008)

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<sup>3</sup> Hong et al (2018) provide experimental evidence that only a subsample of workers in a factory trade off quality for quantity, and Al-Ubaydli et al (2015), Hossain and Li (2014), Englmaier et al (2017), and Kim and Slonim (2021) all find that workers across different settings increase the quality of their work without sacrificing quantity. Chetty et al (2014) find little evidence of tradeoffs between review quality and quicker turnarounds among referees at the Journal of Public Economics. Educational experiments by Glewwe et al (2010) and Muralidharan and Sundararaman (2013) show that performance pay to teachers increases student performance in the incentivized dimensions without adverse effects on unrewarded dimensions. Similarly, Gubler, Liu, and Roman (2021) provide evidence suggesting that emergency crews dynamically adjust their behavior to prioritize both financial and social objectives of their agency.

document large differences in total factor productivity across plants. Research on management practices similarly shows substantial performance dispersion across firms, industries, and countries, suggesting a long tail of establishments that remain far from the productivity frontier (Bloom & Van Reenen, 2007; Bloom et al., 2013; Bruhn et al., 2018).

This distinction matters for goal communication. When employees are on the frontier, making multiple goals salient may heighten tradeoffs. In the context of this experiment, a skilled lineman pair on the productivity frontier may already be conducting the grounding procedure as safely and quickly as possible. For such a pair, reducing time may require sacrificing safety, while increasing safety may require taking more time. When employees are inside the frontier, however, the same communication may improve performance by directing attention and effort toward actions that improve both dimensions. A less skilled pair may reduce time not by skipping safety steps, but by thoroughly assessing safety risks prior to the task to better prepare, coordinate, and think through the steps, enabling them to go through the procedure and use equipment more quickly and smoothly. In this case, communicating both safety and efficiency can help employees identify opportunities for joint improvement that single-goal communication may leave unnoticed.

I therefore expect the effect of communicating multiple goals to vary with employees' baseline performance. Although the productivity frontier is difficult to measure precisely, the existing empirical literature often uses dispersion in performance to infer that lower-performing units are likely to be inside the frontier (e.g., Bloom and Van Reenen 2007). If multiple-goal communication helps employees move toward the frontier, its effects should be largest among pairs that are initially slower and less safe. If employees are already near the frontier, by contrast, communicating multiple goals should be more likely to produce tradeoffs between safety and efficiency.

### **2.3 Goal communication and best practice observation**

The second contingency is the information available to employees to identify how to improve. Firms rarely communicate goals in isolation. They often combine goals with feedback, examples, templates, workshops, personnel circulation, or observations of high performers intended to help employees translate goals into action (Nonaka, 1994; Knott, 2003; Kane, Argote, & Levine, 2005; Almeida & Kogut, 1999; Haas & Hansen, 2007; Jensen & Szulanski, 2008; Szulanski & Jensen, 2008; Lawrence, 2020; Inkpen, 2005, 2008; Bloom et al., 2013). Such observation can help employees improve, but it also creates an inference problem: employees must decide which observed actions are worth copying, why those actions worked, and whether those actions will translate to their own performance.

Goal communication may therefore shape not only what employees try to improve, but also what they take from the information available to them. When employees act directly after feedback, they still need to infer how to improve. But that inference is anchored in their own task experience,

mistakes, and prior performance: they are deciding how to improve what they themselves just did. By contrast, when employees learn from observing others, they must interpret another person's behavior. Unless that observation is explicitly labeled or explained, employees may not know which actions caused high performance, which goals those actions advanced, or whether those actions should be copied in their own execution.

This creates a different role for multiple-goal communication. When one goal is salient, employees can evaluate observed actions primarily in relation to that goal. When multiple goals are made salient, however, employees must infer whether observed actions improve one goal, the other goal, or both, and whether copying those actions would create tradeoffs in their own performance. Multiple-goal communication may therefore expand the set of possible interpretations of the same observed behavior, making it harder to decide what to copy and why.

This logic builds on work showing that interdependent goals and complex performance systems make it difficult to infer which actions produce which outcomes (Ethiraj & Levinthal, 2009; Hu & Bettis, 2018; Rivkin, 2000). It is also consistent with research on causal ambiguity, which suggests that when the link between actions and outcomes is difficult to observe or interpret, decision-makers may draw incorrect inferences about what drives performance and struggle to transfer effective practices across settings (Lippman & Rumelt, 1982; Zander & Kogut, 1995; Ryall, 2009). Finally, it relates to paradox theory, which suggests that when actors approach competing demands as tradeoffs rather than as tensions that can be jointly managed, they may be less able to identify integrative responses (Smith & Lewis, 2011; Miron-Spektor et al., 2018).

Together, this logic suggests that communicating multiple goals may have different effects depending on the information employees use to improve. When employees act directly on feedback and have room to improve on multiple dimensions, multiple-goal communication may improve performance by directing attention and effort toward joint improvement. But when employees learn from unlabeled observation, the same communication may make observed behavior harder to interpret, weakening the gains from observation by increasing ambiguity about what to copy and why.

### **3. SETTING**

The experiment was run in collaboration with a multinational energy corporation, which provided a natural context to explore the impact of communicating multiple goals. Many energy companies increasingly communicate goals other than profits, such as safety or sustainability. Furthermore, these organizational goals often map directly to operational goals that employees carry out at the frontlines of the company, meaning that customer perceptions are often formed in part by how employees carry out these goals in practice.



In this company, safety and efficiency were seen as key operational goals that employees could pursue. As in many other settings (Bennett, 2014; Greve & Seidel, 2014; Gaba & Greve, 2019),<sup>4</sup> safety and efficiency were viewed as being directly in conflict with each other: pursuing safety was seen as taking more time, and thus perceived to come at the cost of efficiency. A manager described: “Although safety is of high importance, sometimes it takes too much time. It might take 45 minutes to comply with the full safety protocol, even though the task might take 5 minutes....this makes it very hard to be efficient.”

The company saw safety as a key goal that it sought to continuously improve on, rather than a constraint to be met. It tracked a range of safety measures beyond the number of accidents and fatalities—such as performance based on specified standards for every procedure and the number of safety violations—and was proud that their safety performance improved over time. Safety was thus seen and expressed as a goal, intended to direct where employees decided to be on the safety-efficiency spectrum. While it may seem intuitive that employees would always prioritize safety given that their own lives were at risk, many appeared to be desensitized to the potential risks of their daily work and often did not take safety precautions when not prompted, especially as accidents were a low-likelihood event. The company thus took great effort to ensure that employees followed even the most basic safety procedures.

At the time of the experiment, there was a broad consensus across the company that safety was a key goal, but internal debate and variation on whether safety alone should be communicated to employees to prioritize, or safety along with efficiency. Some managers stated that safety should never be compromised and “never be under negotiation, regardless of any efficiency goals.” Others stated that while safety was a key goal, it could not come at the expense of efficiency, and advocated for mentioning efficiency as an additional priority.

This variation in managerial goal emphasis provided an opportunity to experimentally and credibly vary goal communication. The randomization of videos identifies treatment effects, but the pre-existing variation in managerial emphasis helped make both messages natural within the organization. Because managers varied in which goals they emphasized, and linemen rotated across managers on a regular basis, employees followed what each manager supervising them at any given time communicated. As one employee voiced, “Every supervisor says different things, so I never know how much I should prioritize safety or efficiency. It really depends on who is supervising me that day.” This made it possible for managers to communicate either safety alone or both safety and efficiency without one message seeming less credible to employees.

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<sup>4</sup> There are many examples across industries suggesting a conflict between safety and efficiency, where managers have sometimes resolved this conflict in favor of profitability. For example, General Motors sold cars with faulty ignition switches for over ten years knowing the grave safety risks they posed, as fixing or replacing them would be costly (Bennett 2014). Airlines investment in safety initiatives such as fleet maintenance and replacement require a significant investment of resources (Gaba and Greve 2019).

## 4. EXPERIMENTAL DESIGN

[INSERT TABLE 1 HERE]

The experiment was run across all 540 linemen across three company subsidiaries as they performed a core procedure that is part of every task on live electrical lines. Linemen worked in pairs, which meant that 270 pairs in total were part of the experiment. Linemen pairs were monitored and assessed on two outcomes: (1) efficiency, measured by the time they took to complete the procedure, and (2) safety, coded using a score based on an internal company rubric. Each pair was randomly assigned to one of four experimental conditions that cross-randomized two treatment arms, one that varied goal communication and another that varied whether the pair observed high-performers. Table 1 summarizes the content of each condition.

### 4.1 Experimental context, sample, and data collection

At the time of the experiment, all linemen pairs were being retrained and evaluated on an operationally important procedure called grounding, a time-consuming and safety-relevant task that enabled a natural study of multiple goals. Grounding consists of connecting electrical lines to the earth to stop any electricity flow. Because it needs to be performed at the beginning and end of any task on live electrical lines, it is a crucial task for all linemen, and has large implications for both efficiency and safety. Potential safety violations can be fatal, and yet if grounding takes two hours, the total time spent on grounding would span four hours, meaning that a linemen pair could only finish a single task in a day's shift.

While safety and efficiency are both important, they are unlikely to have positive complementarities at the productivity frontier. For example, key steps to increase safety involve carefully analyzing all potential risks and testing all equipment prior to beginning the procedure. Doing these steps thoroughly increases the time taken for the procedure, directly reducing efficiency. Conversely, linemen can speed through steps like checking for no voltage or brushing parts to remove rust and contaminants to ensure a good connection, but this gain in speed would risk safety.

Between September 2018 and April 2019, every linemen pair was scheduled for two rounds of grounding evaluations slated three months apart. The first of the evaluations was used to measure baseline performance, and the second as an opportunity to run the experiment. These were communicated as regular training evaluations, which happened multiple times throughout the year for various key procedures.

Each pair came into the training center at each of the subsidiaries at prearranged times, where they were evaluated by a training supervisor. To obtain verifiable measurability of outcomes to rule out any inefficient contracting challenges, linemen were additionally documented by two sets of

cameras, one on the ground providing the same view that the supervisor could see, and another on the linemen's body that provided a detailed view of footwork and the work being performed on the lines.

Supervisors evaluated linemen pairs on two main outcomes. They measured efficiency by recording the time they took to complete each step of the grounding procedure, as well as the full time taken. They measured safety using an internal company rubric that enabled them to assign scores from zero (worst) to five (best) for each step, which precisely outlined how to assign each score based on specific observable behavior and what may be missing. This score totaled to a maximum of twenty-five for the entire procedure. Supervisors directly entered these measures into an online database, and a head supervisor checked over all evaluations and cross-validated them each week, using the recorded linemen footage as reference if necessary. The safety rubric was standard knowledge within the company, and all linemen were fully aware of the details of the rubric and its importance.

These evaluations provided a context with high career stakes, because evaluation outcomes generally affected promotions and status changes. Poor performance resulted in sanctioning of work, as linemen who did not meet set thresholds had to undergo further evaluation before being able to continue working in the field.

The experiment was approved by the relevant IRB offices and registered in the AEA RCT Registry. Linemen were not informed *ex ante* that the second evaluation round differed from standard procedures. However, because randomly assigned interventions were designed to affect linemen performance, individual pair results from the second round were not disclosed to direct managers and were not used for work sanctioning or promotion decisions. All treatment arms added training content rather than withholding the standard feedback provided in the *Control* condition.

## 4.2 Experimental interventions

When linemen arrived at the training center for their second round of evaluations, they were directed to a computer at the training center set up with a feedback video of approximately 1 to 6 minutes to watch prior to performing the grounding evaluation.<sup>5</sup> All videos included the baseline *Control* video, which showed the training supervisors providing safety feedback on the part of the procedure that the pair had performed worst on in their baseline evaluation (first round). The grounding procedure was taught and evaluated at the company as a sequence of five steps, where each step was assessed separately in terms of time taken and safety. To ensure that videos were not too long and that differences in video lengths between treatment conditions were small, the company chose to focus on the worst-performed step for each linemen pair.

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<sup>5</sup> The shortest video was 41 seconds, and the longest 6 minutes and 8 seconds. The *Control* videos ranged from 41s to 1m 58s. (average length 1m 5s); *Multiple Goals* videos ranged from 1m 1s to 2m 19s (average length 1m 26s); *Best Practice* videos ranged from 2m 34s to 5m 48s (average length 3m 45s); and *Best Practice + Multiple Goals* videos ranged from 2m 58s to 6m 8s (average length 4m 5s).

The type of video provided varied depending on the pair's assigned experimental condition. Each lineman pair was randomly assigned to one of four experimental conditions (Table 1), stratified by subsidiary. In this setting, contamination was possible in principle, but unlikely to materially affect the estimates. The timing was central: each pair watched the assigned video and then immediately performed the evaluation, so any conversation after a pair's evaluation could not affect that pair's already-measured outcome. For contamination to matter, a pair would have had to learn treatment-specific content from another pair before watching their own video and completing their own evaluation. This was unlikely because pairs came individually to the training center at assigned times and were otherwise dispatched daily to different worksites across the country.

The *Control* video was the base video, on top of which treatments were layered. This video showed a supervisor who emphasized the importance of safety and provided feedback on how to improve it, beginning as follows:

“Hello, my name is [Supervisor's Name], and I am here to provide feedback on your performance during the evaluation of the grounding procedure. The entire procedure involved multiple steps, but I will focus on the specific step where you have the greatest opportunity for improvement. I would like to talk about step [Number], [Name of Step], and the importance of safety.”

The supervisor then elaborated on what prioritizing safety meant for this step, which varied depending on how linemen performed on that step at baseline. For example:

“When a change in the work plan introducing significant risks occurs, the task should be stopped to review the hazards according to the Risk Control Hierarchy and indicate any controls and changes in the job safety analysis before continuing with the work in progress.”

All videos ended with: “Thank you for viewing this feedback session. Please consider all of this information when performing the second round of evaluations” (Online Appendix A shows scripts of videos).

Each treatment arm layered an additional 1 to 5 minutes of video material onto the *Control* video. The first treatment arm, *Multiple Goals*, extended the *Control* video with the supervisor additionally emphasizing the importance of prioritizing efficiency. The supervisor began with, “I would also like to talk about the importance of efficiency,” and elaborated that while safety is important, they must also execute the step “in the shortest possible time.” There were 5 versions of this treatment, one for each feedback step, to emphasize this goal of efficiency within the context of the step that linemen received feedback on. For example, one of the messages stated:

“I would also like to talk about the important of efficiency. Grounding removal remains an important and critical part of efficient task execution, and delays due to improper handling of equipment and lack of task planning before ascending the pole should be avoided.”

The second treatment arm, *Best Practice*, enabled the observation of best practices that linemen could potentially learn from. It layered onto the *Control* or *Multiple Goals* videos an additional video showing a high-performing linemen pair from another subsidiary performing the grounding step, following a common way in which best practices are transferred within companies (Argote & Fahrenkopf, 2016; Inkpen, 2005; 2008). To create these best practice videos, training supervisors chose footage among linemen that displayed best practices for each of the five steps. All linemen were shown a video from a different subsidiary, to reduce the possibility that these videos may trigger peer effects that could affect motivation. The videos included no instruction or verbal audio of any kind, and simply enabled the observation of linemen. It displayed a short text at the beginning of the video to signpost the step that was being executed (screenshots shown in Appendix Figure A.2), but there was no other communication provided or explanations of how the practices related to safety or efficiency.

[INSERT TABLE 2 HERE]

These two treatment arms were cross-randomized to result in four experimental conditions: *Control*, *Multiple Goals*, *Best Practice*, and *Best Practice + Multiple Goals*. The videos were combined so that every linemen pair watched only a single video. After watching their assigned video, linemen pairs performed their second round of grounding evaluations. Table 2 shows that randomization produced balanced samples across experimental conditions, with no economically or statistically significant differences across linemen on observable characteristics. There was no attrition or non-compliance.

Each subsidiary had its own training center and set of training supervisors (a total of nine across the three subsidiaries), and each linemen pair was assigned to the training supervisor in their subsidiary who was scheduled for the day that they were assigned to in each round. Training supervisors were blind to experimental conditions and details. Each pair had a unique video number assigned to view inside the training center, and evaluations were conducted outside on training grounds.

## 5. ECONOMETRIC MODEL

To evaluate the impact of communicating multiple goals on linemen performance, I estimate average treatment effects using the following regression model:

$$Y_i = \beta_M \text{Multiple Goals}_i + \beta_B \text{Best Practice}_i + \beta_{BM} \text{Best Practice}_i * \text{Multiple Goals}_i + \gamma Y_{i,t=0} + \delta_i + \varepsilon_i$$

$Y_i$  represents the outcomes of interest for linemen pair  $i$ , which include the time taken and the overall safety score for both the full procedure and the specific step on which they were given feedback.  $Y_{i,t=0}$  controls for the baseline value of the outcome variable, and  $\delta_i$  is a randomization stratum indicator -- neither of which are necessary for identification but help absorb noise.  $\varepsilon_i$  is an idiosyncratic error term.

In this model,  $\beta_M$ ,  $\beta_B$  and  $\beta_{BM}$  are the coefficients of interest.  $\beta_M$  estimates the effect of the *Multiple Goals* treatment compared to the Control, evaluating the impact of communicating safety and efficiency compared to communicating safety alone.  $\beta_B$  identifies the impact of adding unlabeled best-practice observation to the *Control* video.  $\beta_{BM}$  estimates the differential effect of adding the same unlabeled observation to the *Multiple Goals* video, identifying whether communicating multiple goals changes what employees take from the same observed behavior.

I additionally assess the robustness of results across alternative specifications, including a simple comparison in means, as well as a difference-in-differences model. Standard errors are clustered at the pair level, the unit of randomization, for all specifications (Abadie et al., 2017).

## 6. GOAL COMMUNICATION AND PERFORMANCE

At baseline, linemen pairs showed a large dispersion in efficiency and safety, with a long tail of pairs that were both unsafe and inefficient. This suggests that at least some pairs were inside the productivity frontier with room to improve on both dimensions.

After treatment, linemen pairs assigned to *Multiple Goals* improved efficiency without sacrificing safety compared to those assigned to Control. Consistent with the productivity-frontier interpretation, these gains were largest among pairs with more room to improve. *Best Practice* also improved efficiency and safety on average, suggesting that observation can help employees improve. However, when the same unlabeled observation was paired with multiple-goal communication, the gains from either intervention weakened.

### 6.1 Baseline variation in efficiency and safety

Linemen pairs showed a large dispersion in baseline efficiency and safety. The variation in efficiency ranged from thirty minutes to two hours, implying that the fastest pairs took roughly a third of the time taken by the slowest pairs (Figure 2(a)). Safety scores ranged from 12 to 25, indicating that more than 10% of linemen failed to meet the minimum safety criteria of 15 required to continue working as a fully-functioning employee (Figure 2(b)).

[INSERT FIGURE 2 HERE]

Notably, not all slow pairs were safe, and not all unsafe pairs were fast. Plotting every pair by baseline time and safety score reveals three groups: pairs who were slower and less safe compared to the median, pairs who were safer but slower or faster but less safe, and pairs who were both safer and faster (Figure 2(c)). There is also one pair that achieved both the highest safety score of 25 and the fastest time of 29.92 minutes, suggesting that other pairs may be inside the productivity frontier. This is consistent with growing evidence that many firms may be far from the productivity frontier (Syverson, 2004; Bloom & Van Reenen, 2007; Bloom et al., 2013), and provides complementary evidence of dispersion in practices even within a large multinational firm.

Nevertheless, there is direct evidence that safety and efficiency trade off at the margin. Baseline time and safety are negatively correlated at both the pair level (Pearson  $r=-0.176$ ,  $p=0.004$ ; Spearman  $\rho=-0.164$ ,  $p=0.007$ ) and the step level ( $r=-0.097$ ,  $p<0.001$  across 1,350 step-observations). These correlations are negative but modest, consistent with the idea that safety and efficiency can trade off in some cases while many pairs still have room to improve on both dimensions. There is also evidence of tradeoffs, defined as cases in which a pair improves in one dimension while deteriorating in the other: either becoming faster but less safe, or becoming safer but slower. Across the sample, 32% of linemen pairs show evidence of such a tradeoff across rounds, and 37% of pairs trade off between the two dimensions in the *Control* condition (Appendix Figure 5). Consistent with this interpretation, additional data identify safety-relevant actions that pairs skipped or executed incompletely, showing the concrete safety steps linemen sometimes forgo to reduce time (Appendix Table 7).<sup>6</sup> This evidence mirrored the broader view articulated by most employees, who highlighted the tradeoffs they believed were present. For example, one lineman explained, “Taking safety steps takes more time. The job safety analysis, and considering all risks takes time. All of this adds to the task time.”

## 6.2 The impact of communicating multiple goals

[INSERT TABLE 3 AND FIGURE 3 HERE]

Compared to linemen in the *Control* group, linemen pairs assigned to *Multiple Goals* showed higher efficiency. Figure 3 shows that the *Multiple Goals* treatment not only improved average efficiency, but also shifted the full distribution of linemen to the left, decreasing the time they took.

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<sup>6</sup> For example, Step 1 requires a comprehensive Job Safety Analysis (JSA), including the sequence of tasks, work hazards, hazard-control actions, and an emergency plan. Skipping the JSA or completing it only partially reduces the time required to complete grounding, but reduces safety from the company’s perspective. In Round 1, 0.37% of pairs entirely skipped the JSA, 2.22% skipped risk or hazard analysis, 39.26% skipped some fields in the protocol document or did not sign the JSA, and 98.52% fell below the full documentation standard.

The first two rows of Table 3 columns (1) - (2) indicate that this improvement was an 11% decrease ( $p=0.001$ ) on average compared to the *Control* group, who took 64 minutes.

[INSERT FIGURE 4 HERE]

This improvement in efficiency did not come at a significant cost to safety. The treatment effect estimate of *Multiple Goals* in Table 3 suggests that linemen also received higher safety scores by 1%, which is imprecisely estimated ( $p=0.56$ ). Figure 4 shows that the majority of linemen improved in both efficiency and safety, contrary to theoretical predictions that multiple goals will result in confusion and reduce performance (Jensen, 2002). Given that safety and efficiency are unlikely to have a natural positive complementarity at the frontier, this result, together with the baseline dispersion and tradeoff evidence above, raises the possibility that the effect of multiple goals may depend on where employees are relative to the productivity frontier.

[INSERT TABLE 4 AND HERE]

Consistent with the interpretation that multiple-goal communication helps most when pairs have room to improve, linemen assigned to *Multiple Goals* who were slower and less safe at baseline were more likely to improve on both efficiency and safety (Figure 5). These pairs were 23 percentage points more likely to improve on both dimensions when assigned to *Multiple Goals* ( $p = 0.039$ ), a larger and more precise estimate than for the other baseline groups (Table 4). Splitting the outcome by dimension shows that *Multiple Goals* reduced log time among slower-and-less-safe pairs ( $-0.15$ ,  $p = 0.024$ ) and off-diagonal pairs ( $-0.15$ ,  $p = 0.004$ ), but not among safer-and-faster pairs ( $-0.02$ ,  $p = 0.781$ ), with no reduction in safety in any group (Appendix Table 8). Appendix Figure 6 also provides a more direct approximation of distance from the empirical frontier, by measuring each pair's standardized baseline safety and time relative to the best observed baseline performance: a safety score of 25 out of 25 and a completion time of 29.92 minutes. Although this frontier is proxied by a single observed extreme point rather than estimated structurally, pairs farther from the empirical frontier improve more than pairs closer to it. Together, these analyses support the interpretation that communicating multiple goals can improve performance by directing attention and effort toward efficiency as well as safety when employees are inside the productivity frontier and have room to improve on both dimensions.

### 6.3 Communicating multiple goals with best practice observation

[INSERT FIGURE 6 HERE]



*Best Practice* also improved performance on average, consistent with evidence of positive productivity spillovers from high-performing peers in other settings (e.g., Mas and Moretti, 2009; Blanes i Vidal and Nossol, 2011). Figure 6 shows that the entire distribution shifted to the left for efficiency, decreasing the time pairs took, and shifted to the right for safety. Table 3 Columns (1)-(2) and (4)-(5) show a 7% improvement in efficiency (~4min;  $p=0.046$ ) and a 3% improvement in safety (0.6 points;  $p=0.026$ ) on average.

Improvements from *Best Practice* were larger and more precisely estimated for pairs that were safer and faster at baseline. These pairs assigned to *Best Practice* were 49 percentage points ( $p=0.043$ ) more likely to improve in both safety and efficiency relative to those assigned to *Control*. That said, this pattern remains broadly consistent with the productivity frontier interpretation. The safer-and-faster baseline quadrant is defined relative to the baseline median splits, rather than the frontier itself. Within this group, we also observe that gains are concentrated among pairs farther from the empirical frontier, reducing time by 0.19 log points and increasing safety by 0.10 log points (Appendix Table 9). The corresponding estimates for pairs nearer the frontier are small and not statistically significant. This suggests that even some safer-and-faster pairs still had room to improve, while those closer to the frontier improved less. Together, these findings are consistent with the interpretation that many linemen operate inside the productivity frontier, leaving room to improve on both safety and efficiency. At the same time, the results suggest that pairs may differ not only in how much room they have to improve, but also in their ability to assimilate and apply what they observe from high-performing peers, consistent with the broader absorptive-capacity literature (Cohen and Levinthal 1990, Zahra and George 2002).

[INSERT FIGURE 7 HERE]

Despite the positive effects of both treatments when assigned on their own, performance gains weakened when they were combined in *Best Practice + Multiple Goals*. Figure 7 shows that compared to linemen who received *Best Practice* alone, linemen assigned to *Best Practice + Multiple Goals* took longer and received lower safety scores across the distribution. They were also slower and less safe compared to linemen assigned to *Multiple Goals* alone, with efficiency reductions along the distribution. These represented an efficiency decrease of 23% (13min;  $p<0.001$ ), and a safety deterioration of 4% (0.6 points;  $p=0.078$ ) (Table 3), weakening the gains from either treatment when implemented alone. There was little evidence of significant heterogeneity in this combined treatment effect (Table 4), suggesting that what weakened these gains may have applied similarly across pairs.

[INSERT FIGURE 8 HERE]

Together, these findings suggest that goal communication affects performance by directing what employees attend to and allocate effort on. Communicating multiple goals improves performance when employees act on feedback and have room to improve on both dimensions. But when paired with best practice observation, the same communication weakens the performance gains from observation. These results are robust to different econometric models (Figure 8, Appendix Table 4), as well as to evaluating linemen performance on the specific step on which they received feedback (Appendix Table 1) and including step and supervisor fixed effects (Appendix Table 2).

## **7. WHY DOES COMMUNICATING MULTIPLE GOALS WEAKEN GAINS FROM BEST PRACTICE OBSERVATION?**

The findings described in the previous section indicate that communicating multiple goals improved performance when provided alone, but weakened gains when paired with best practice observation. The natural question this raises is why. While I cannot precisely identify the mechanism, the design of the field experiment and contextual evidence help narrow the plausible explanations.

First, any explanation relating to the potentially negative effects of the best practice video alone can be ruled out given the performance-enhancing impact of the *Best Practice* treatment on its own. For example, if this effect were driven by the fact that the best practice video was unclear or poor in quality, then the effects of *Best Practice* alone should have also been negative, but in contrast, it improved both linemen's safety and efficiency.

Similarly, the result cannot be explained by multiple-goal communication being harmful on its own. A common explanation for negative performance effects from multiple goals is direct tradeoffs between goals (Etzioni, 1964; Holmstrom & Milgrom, 1991; Meyer, 2002; Zbaracki & Bergen, 2010). A related explanation is that goal interdependence makes it difficult for employees to know which actions lead to performance improvements (e.g. Ethiraj & Levinthal, 2009). However, these explanations are inconsistent with the positive effects of *Multiple Goals* when assigned alone, which improved efficiency without reducing safety.

Third, given that all videos focused on the step that the linemen pair performed worst on at baseline, it could be that the videos assigned in *Best Practice + Multiple Goals* compared to the other conditions happened to target more difficult steps. However, as expected from randomization and the balance check, the distribution of feedback steps across experimental conditions was not statistically or economically significantly different (Appendix Figure 4). Furthermore, the results are robust to controlling for the step shown in the video (Appendix Table 2).

Fourth, it could be that the *Best Practice + Multiple Goals* treatment was too long or incoherent compared to the other videos. While this is possible, it seems unlikely given the context. First, the *Best Practice + Multiple Goals* videos were on average 21 seconds longer than the *Best Practice* videos. Second, the *Best Practice + Multiple Goals* videos simply combined the *Multiple Goals* and *Best*

*Practice* videos, and all videos were vetted by multiple levels across the company. The head of the training team along with two of his team members, the head of all field operations, the head of safety, Director of Planning and Engineering, and the Vice President of Operations all oversaw the process and vetted them as appropriate training assets to help linemen improve. Many of these individuals had been involved in training and development for several years and had previously worked as linemen, providing them with deep knowledge and experience. Moreover, linemen assigned to *Best Practice + Multiple Goals* did not display a lower likelihood of executing the correct sequence of steps as highlighted in the video, suggesting that they were not less likely to attend to the information provided (Appendix Table 5).

This leaves explanations related to how multiple-goal communication changes what employees take from best practice observation. One possibility is that multiple-goal communication creates different kinds of inference problems depending on the information employees receive with it. When employees act directly after feedback, communicating multiple goals directs attention to the dimensions on which they can improve their own performance. Although employees still need to identify specific actions that improve safety and efficiency, this inference is anchored in their own task experience and prior performance.

By contrast, learning from best practice observation requires employees to interpret another pair's behavior without labels explaining which observed actions matter, which goals those actions advance, or whether those actions would translate to their own performance. In this setting, making multiple goals salient expands the set of possible interpretations of the same observed behavior. Employees must infer not only what to copy, but also why it worked, whether it improved safety, efficiency, or both, and whether copying it would create tradeoffs in their own execution. Multiple-goal communication may therefore improve direct performance while weakening learning from observation because it increases ambiguity about what to copy and why.

This logic builds on prior work showing that interdependent goals and complex performance systems make it difficult to infer which actions produce which outcomes (Ethiraj and Levinthal, 2009; Hu and Bettis, 2018; Rivkin, 2000). It is also consistent with research on causal ambiguity, which suggests that when the link between actions and outcomes is difficult to observe or interpret, decision-makers may draw incorrect inferences about what drives performance and struggle to transfer effective practices across settings (Lippman and Rumelt, 1982; Zander and Kogut, 1995; Zollo and Reuer, 2010; Gupta et al., 2015; Ryall, 2009). Finally, this interpretation is consistent with paradox theory, which suggests that when actors approach competing demands as tradeoffs rather than as tensions that can be jointly managed, they may be less able to identify integrative responses (Smith and Lewis, 2011; Miron-Spektor et al., 2018).

While evidence from the experiment cannot precisely pin down the underlying mechanism, information gathered in the field provides suggestive evidence consistent with this interpretation.

Linemen pairs assigned to *Best Practice + Multiple Goals* uniquely commented that they did not find the video helpful, with no similar comments from pairs assigned to other conditions. These pairs elaborated that it was not clear how to determine from the best practice videos what they might do to improve on both goals, as certain actions could improve one goal while worsening the other. One comment echoed by others described, “I observed some things from the best practice video that I could do, but didn’t know if I should try it because it could increase safety but make me slower...so I didn’t know what to change from what I was doing.” Another explained, “I don’t know what to take away from this best practice video, because anything I try won’t help me get both safer and faster. I don’t know how everything is going to balance out in the end.” Some linemen pairs also expressed that clear and concrete guidance on steps would be helpful in improving their procedure. While speculative, this qualitative evidence paired with the experimental findings suggests that multiple-goal communication may have complicated the processing of information, making it harder for employees to decide what to copy from best practice observation.

Conversations with managers one year after the experiment provided consistent insights suggesting that the effects from training may not have been short-term. The head supervisor commented, “To me it is clear that linemen improve[d] their skills or did not...as a result of training. [...] Performance in the second round showed what linemen could do as a result of more [and different] training, which did not change outside the training center.” He elaborated that it was difficult for those who had not performed well because “You can only be more efficient...and be safer [when] you have the proper training and execute [that] so many times as possible,” and highlighted that these linemen had to be brought back into the training center to be retrained. Employees were also monitored during their daily work as in the training center, which may have contributed to consistent behavior inside and outside the training center.

Consistent with these observations, the company did not move forward with the best practice observation in their training, and moved forward by communicating both safety and efficiency. They implemented a different way to transfer best practices by having supervisors demonstrate every detailed step slowly while providing specific instructions.

## **8. EXTERNAL VALIDITY**

The key motivation behind this experiment was to establish a high degree of internal validity to provide a test of the causal effects of communicating multiple goals (List 2020). In this sense, this context provides a unique opportunity within a large energy corporation, where single-goal and multiple-goal communication can both be communicated credibly and naturally, randomization can be ensured without non-compliance, and both outcomes can be measured precisely with full monitoring to limit measurability issues. However, these features also make the context particular. While these findings are more likely to be applicable to similar settings involving frontline workers

who carry out activities with multidimensional objectives, how far they apply is an open question for future work.

Three key criteria provide insight on how generalizable a field experiment may be (List 2020): (1) the selection of the sample that may affect responses to treatment; (2) the degree of attrition and non-compliance suggesting issues that could limit compliance in other settings; (3) the naturalness of the choice task, setting, and time frame that influences how interchangeable the nature of choice is between the research setting and target settings.<sup>7</sup>

On selection, there are three important factors to consider. First, this study examines pairs of linemen, which may provide different results than for individuals or larger teams. Larger units may have more difficulty coordinating efforts and learning, meaning that improvements in both goals may be harder to obtain across teams larger than pairs, and the weakening of gains from combining multiple-goal communication with unlabeled observation may be even larger. Second, the study examines the goals of safety and efficiency. Safety and efficiency may differ from other goals, and goal communication may also work differently when goals are less measurable. Moreover, the findings may not generalize to more than two goals. Third, these findings are most indicative of how communicating multiple goals affects employee performance, rather than organizational performance.

On the latter two criteria, the study suggests a higher level of external validity. There is no attrition or non-compliance, and the nature of choice is nearly unchanged between the research setting and target settings. Employees in their regular work evaluations are provided with communication on multiple goals, and make decisions with no artificial constraints imposed, where their outcomes are simply observed and recorded. An important consideration is that this experiment occurred in a training context rather than everyday work. While in this company context, monitoring and the stakes involved in the training evaluation were similar to everyday work, this may have different effects in other settings: it might decrease the magnitude of effects in cases where training evaluations do not have high stakes, or increase the magnitude in cases where everyday work does not involve monitoring or incentives. Furthermore, more precise measurement of longer-term outcomes may provide different insights.

## **9. DISCUSSION**

This paper explores how communicating multiple goals affects employee performance, using a field experiment that varied goal communication to frontline employees in an energy corporation. The findings suggest that communicating multiple goals can improve performance when employees have

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<sup>7</sup> List raises a fourth criteria for programmatic studies targeted to policymakers: what the factors would be that would affect who should receive the program, how it should be implemented, and whether this would pass a benefit vs. cost test if implemented at scale. Given that this last criteria does not apply to this study, which does not propose an intervention for policymakers, I assess the study according to each of the first three criteria.

room to improve on both dimensions inside the productivity frontier: linemen pairs became faster without becoming less safe. The results also show that the same communication can weaken gains when paired with best practice observation, suggesting that goal communication shapes what employees attend to when deciding what to copy from observed behavior.

These results stem from a particular company, and are most likely to generalize across similar contexts involving frontline workers who carry out activities with multidimensional objectives. The results from this paper suggest that the positive effect of multiple-goal communication may be less likely in settings where employees are already close to the frontier, where the goal dimensions involve harder tradeoffs, or where goal communication is paired with unlabeled observation rather than explicit instruction. It may also be that the impacts of introducing new goals to the organization may be different from the effects of communicating already-existing goals. A fruitful area for future work may be to explore the extent to which these results generalize across a broader set of contexts and over a longer time period.

These findings point to two managerial implications. First, in determining whether to communicate multiple goals, managers need to consider whether employees have room to improve on multiple dimensions. Communicating multiple goals may be especially useful when employees are inside the productivity frontier and can improve on one goal without sacrificing another. Second, managers need to consider what other information accompanies goal communication. Best practice information may not always be useful when paired with multiple-goal communication. While firms increasingly have access to data, examples, and observed practices from inside and outside the organization, these findings suggest that employees may need explicit guidance about which observed actions matter for which goals. Without that guidance, communicating multiple goals can weaken the gains from observation.

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**TABLE 1:** Experimental conditions

|                                     | <b>Single-goal communication</b>                                                                                                   | <b>Multiple-goal communication</b>                                                                                             |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>No best practice observation</b> | <b>Control:</b> Supervisor feedback on the pair’s weakest grounding step, communicating safety as the focal goal.                  | <b>Multiple Goals:</b> The <i>Control</i> video plus additional supervisor feedback communicating efficiency as a second goal. |
| <b>Best practice observation</b>    | <b>Best Practice:</b> The <i>Control</i> video followed by an unlabeled video of high-performing linemen completing the same step. | <b>Multiple Goals + Best Practice:</b> The Multiple Goals video followed by the same unlabeled best-practice video.            |

This table describes each of the four experimental conditions. The best practice video was identical across the two best practice conditions and contained no labels, audio instruction, or explanation of which observed actions improve safety, efficiency, or both. This design allows us to compare the effect of communicating multiple goals when employees act directly on feedback, and to test whether communicating multiple goals changes what employees take from the same best practice observation.

**TABLE 2:** Descriptive statistics and balance of variables

| <b><i>Panel A: Descriptive Statistics</i></b> |                  |                  |                  |                                |        |       |
|-----------------------------------------------|------------------|------------------|------------------|--------------------------------|--------|-------|
|                                               | mean             | sd               | min              | p50                            | max    | count |
| Baseline Time Taken                           | 66.28            | 16.21            | 29.92            | 64.30                          | 118.68 | 270   |
| Baseline Safety Score                         | 17.63            | 2.42             | 12.00            | 18.00                          | 25.00  | 270   |
| Baseline Correct Sequence                     | 0.75             | 0.43             | 0.00             | 1.00                           | 1.00   | 270   |
| Baseline Feedback Given                       | 0.34             | 0.47             | 0.00             | 0.00                           | 1.00   | 270   |
| <b><i>Panel B: Balance of Variables</i></b>   |                  |                  |                  |                                |        |       |
|                                               | Control          | Multiple Goals   | Best Practice    | Best Practice + Multiple Goals |        |       |
| Baseline ln(Time Taken)                       | 4.17<br>(0.03)   | 4.12<br>(0.03)   | 4.18<br>(0.03)   | 4.18<br>(0.03)                 |        |       |
| Baseline ln(Safety Score)                     | 2.86<br>(0.02)   | 2.85<br>(0.02)   | 2.88<br>(0.02)   | 2.86<br>(0.02)                 |        |       |
| Baseline Correct Sequence                     | 0.75<br>(0.05)   | 0.75<br>(0.05)   | 0.78<br>(0.05)   | 0.72<br>(0.05)                 |        |       |
| Baseline Feedback Given                       | 0.40<br>(0.06)   | 0.33<br>(0.06)   | 0.31<br>(0.06)   | 0.32<br>(0.06)                 |        |       |
| Days Since Baseline                           | 125.96<br>(3.27) | 123.64<br>(4.04) | 126.52<br>(3.66) | 123.60<br>(3.39)               |        |       |
| Feedback Step                                 | 3.40<br>(0.15)   | 3.54<br>(0.13)   | 3.30<br>(0.16)   | 3.43<br>(0.15)                 |        |       |
| F-statistic p-value                           |                  | 0.63             | 0.95             | 0.96                           |        |       |
| Observations                                  | 68               | 67               | 67               | 68                             |        |       |

Panel (A) provides summary statistics on all baseline characteristics of linemen pairs. Time Taken takes the natural log of the amount of time taken by linemen to finish the grounding procedure at baseline. Safety Score takes a natural log of a measure of the degree to which linemen executed the procedure safely. Correct Sequence indicates whether the procedure sequence was executed correctly. Feedback given indicates whether any immediate feedback was provided in the first round. Panel (b) shows the balance of variables at baseline between experimental conditions, displaying the mean of each variable by condition and standard errors in parentheses. The F-statistic p-value displays the p-value of the F-test for joint significance comparing each treatment separately to the control.

**TABLE 3:** Average treatment effects

|                                | (1)        | (2)            | (3)          | (4)              |
|--------------------------------|------------|----------------|--------------|------------------|
|                                | Time Taken | ln(Time Taken) | Safety Score | ln(Safety Score) |
| Multiple Goals                 | -7.02      | -0.11          | 0.14         | 0.01             |
|                                | (2.09)     | (0.03)         | (0.27)       | (0.02)           |
| Best Practice                  | -4.09      | -0.07          | 0.57         | 0.03             |
|                                | (2.20)     | (0.03)         | (0.25)       | (0.01)           |
| Best Practice + Multiple Goals | 12.99      | 0.21           | -0.62        | -0.04            |
|                                | (3.13)     | (0.05)         | (0.37)       | (0.02)           |
| Strata FE                      | Yes        | Yes            | Yes          | Yes              |
| Controls                       | Yes        | Yes            | Yes          | Yes              |
| Observations                   | 270        | 270            | 270          | 270              |
| Mean (control)                 | 63.79      | 4.13           | 19.00        | 2.94             |
| SD (control)                   | 14.99      | 0.23           | 1.66         | 0.10             |

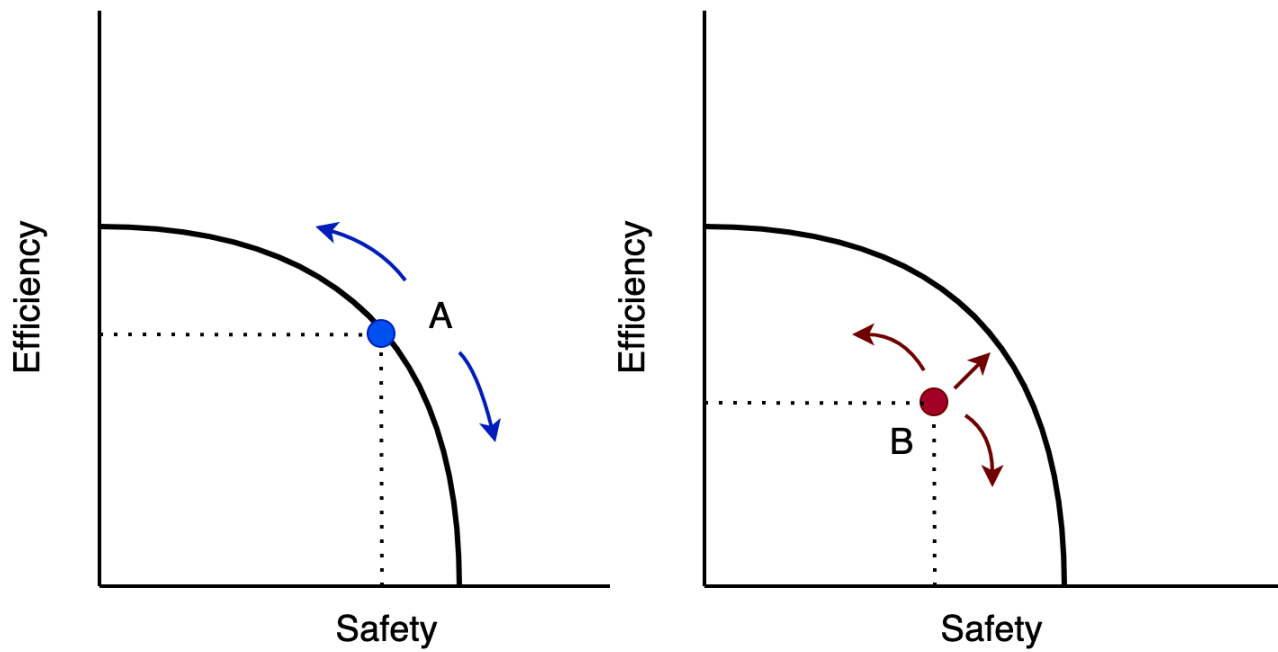
This table shows intention to treat estimates relative to the *Control* group. Observations are at the linemen pair level. The dependent variable for each regression is indicated at the top of the column: *Time Taken* measures the total number of minutes taken to complete the procedure; *Safety Score* is the assigned score using the company's internal rubric. *Controls* include the baseline value of the outcome variable.

**TABLE 4:** Improvement in both efficiency & safety by baseline outcomes

|                                | Improvement in Both Efficiency and Safety |                                             |                          |
|--------------------------------|-------------------------------------------|---------------------------------------------|--------------------------|
|                                | (1) Safer and Faster                      | (2) Safer but Slow, or Faster but Less Safe | (3) Slower and Less Safe |
| Multiple Goals                 | 0.01                                      | 0.12                                        | 0.23                     |
|                                | (0.16)                                    | (0.13)                                      | (0.11)                   |
| Best Practice                  | 0.49                                      | 0.11                                        | 0.09                     |
|                                | (0.16)                                    | (0.13)                                      | (0.14)                   |
| Best Practice + Multiple Goals | -0.48                                     | -0.09                                       | -0.41                    |
|                                | (0.23)                                    | (0.18)                                      | (0.18)                   |
| Constant                       | 0.22                                      | 0.35                                        | 0.74                     |
|                                | (0.15)                                    | (0.11)                                      | (0.11)                   |
| Strata FE                      | Yes                                       | Yes                                         | Yes                      |
| Observations                   | 65                                        | 122                                         | 83                       |

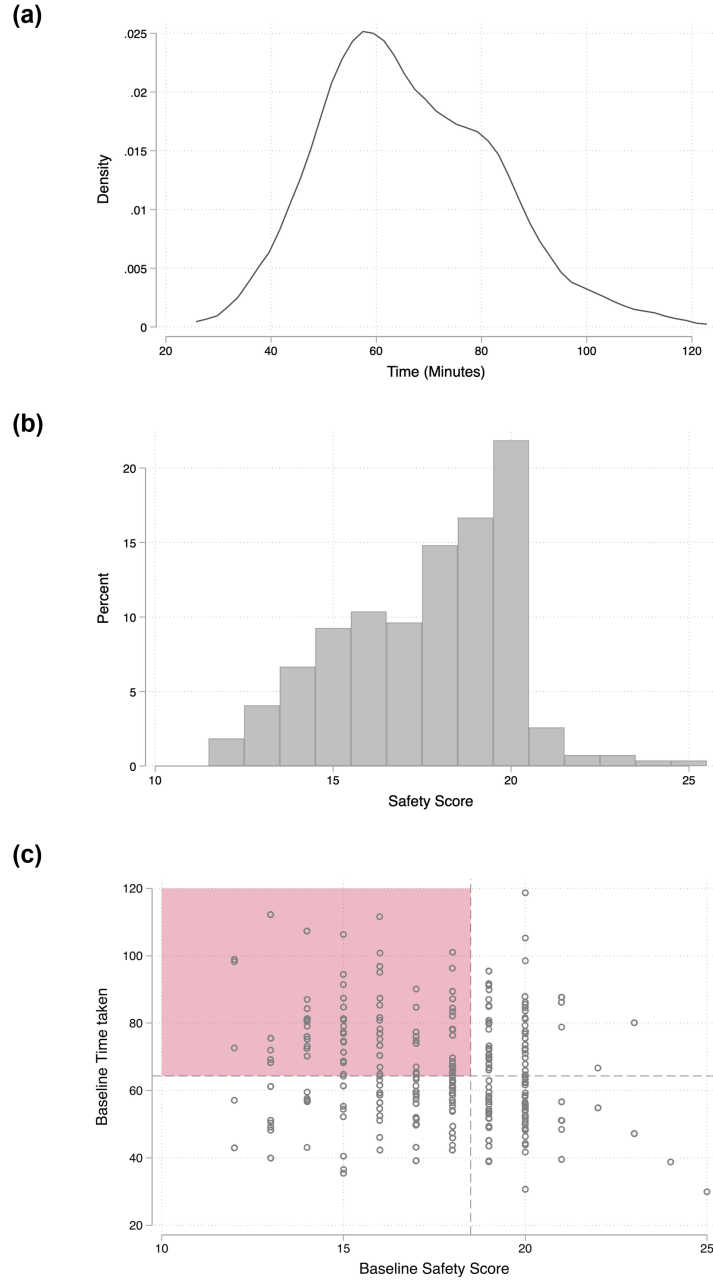
This table shows intention to treat estimates by subgroups based on the linemen pair's baseline quadrant. Observations are at the pair level. The dependent variable for all regressions is a binary indicator for whether both efficiency and safety improved (i.e. equal to or greater than zero). Column 1 shows results for linemen who were safer and faster relative to the median at baseline; Column 2 shows results for linemen who were in one of two baseline quadrants: safer but slow, or faster but less safe, both relative to the median; Column 3 shows results for linemen who were slower and less safe relative to the median at baseline. All regressions control for randomization strata fixed effects.

**FIGURE 1:** Tradeoffs on vs. inside the productivity frontier



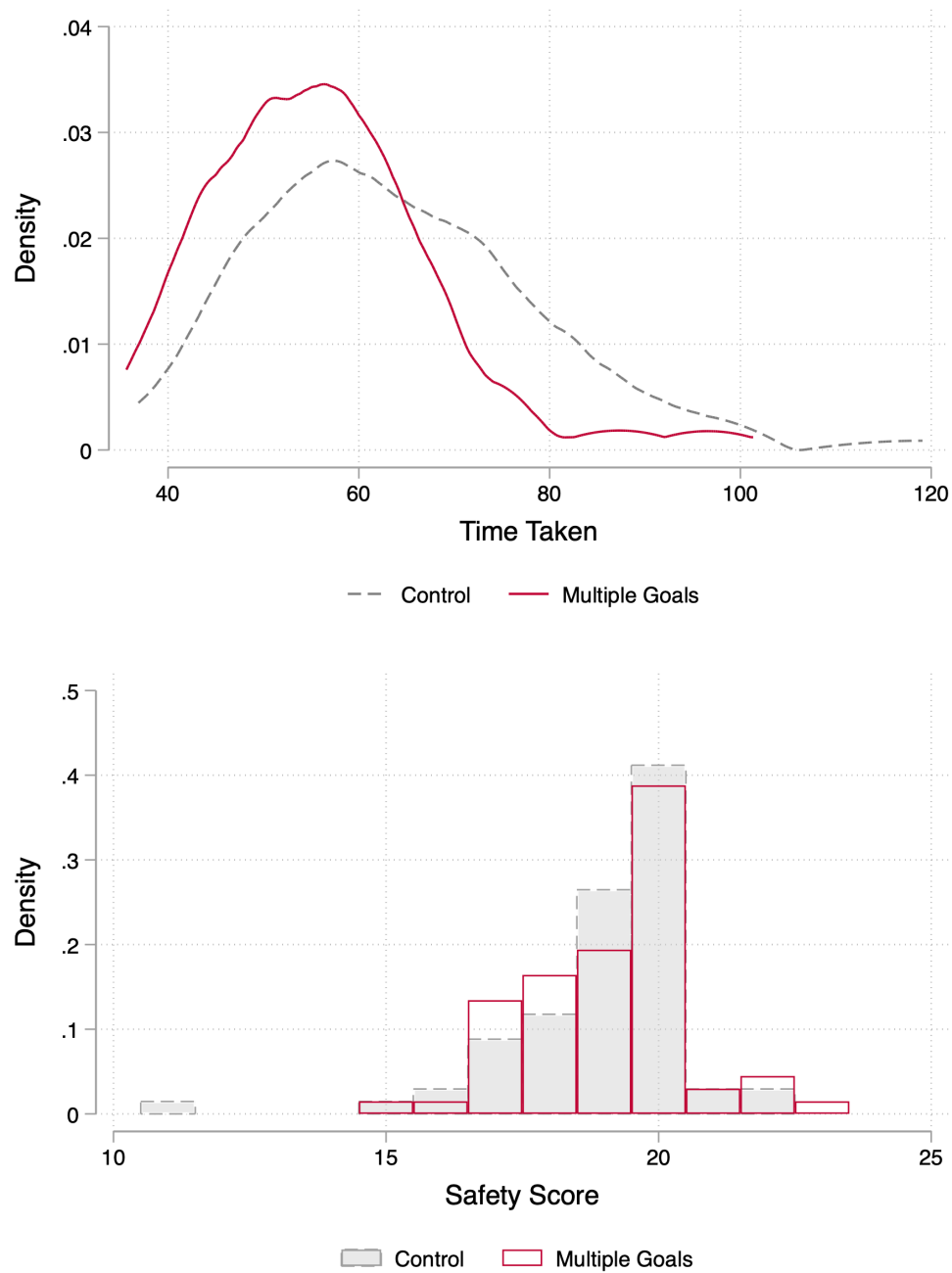
This figure sketches a productivity frontier, where A shows a point on the frontier, and B indicates a point inside the frontier.

**FIGURE 2: Baseline variation in time and safety**



The top two figures (a) and (b) plot the baseline distribution of linemen pairs' times and safety scores for the grounding procedure. The bottom figure (c) plots the relationship between time and safety. Specifically, (c) plots every lineman pair, highlighting four quadrants: those that are less efficient and less safe at baseline compared to the median (top left, highlighted); those that are safer but less efficient, or more efficient but less safe (top right and bottom left), and those that are safer and more efficient compared to the median (bottom right).

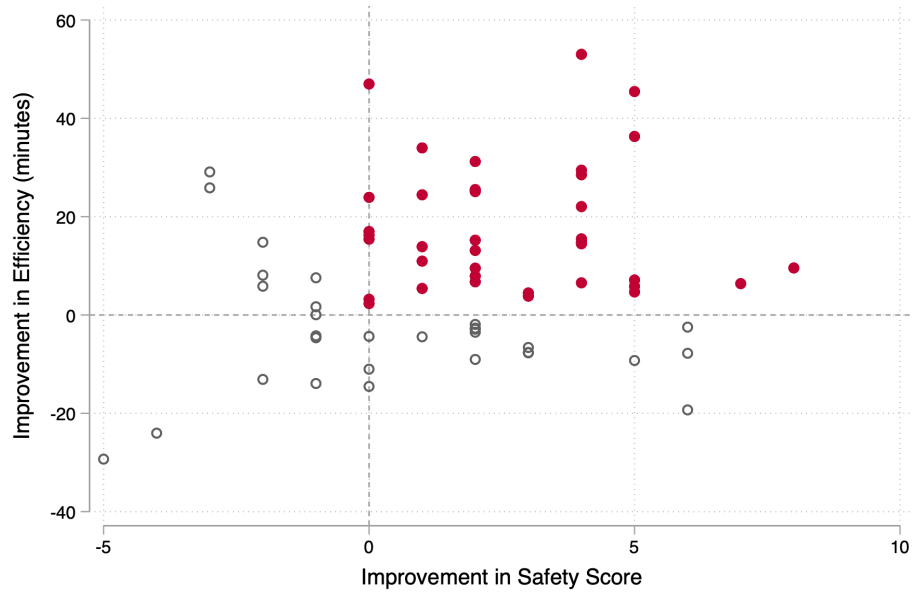
**FIGURE 3:** Average treatment effects of communicating multiple goals



These figures plot the distribution of linemen pairs' times and safety scores in *Multiple Goals* compared to those in the *Control* group. Two-sample Kolmogorov-Smirnov tests for equality of distribution: Time: 0.004; Safety: 1.0.

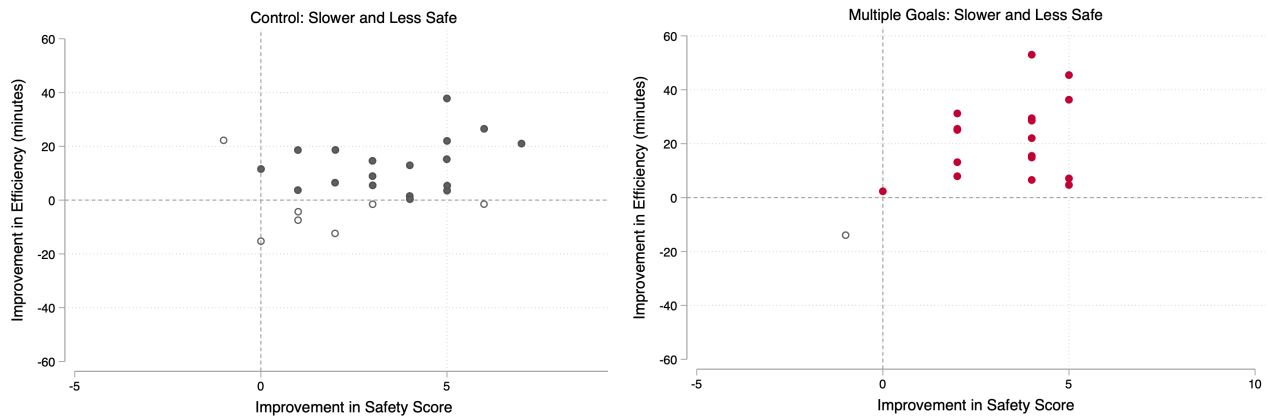


**FIGURE 4:** Improvements in safety and efficiency across pairs assigned to *Multiple Goals*

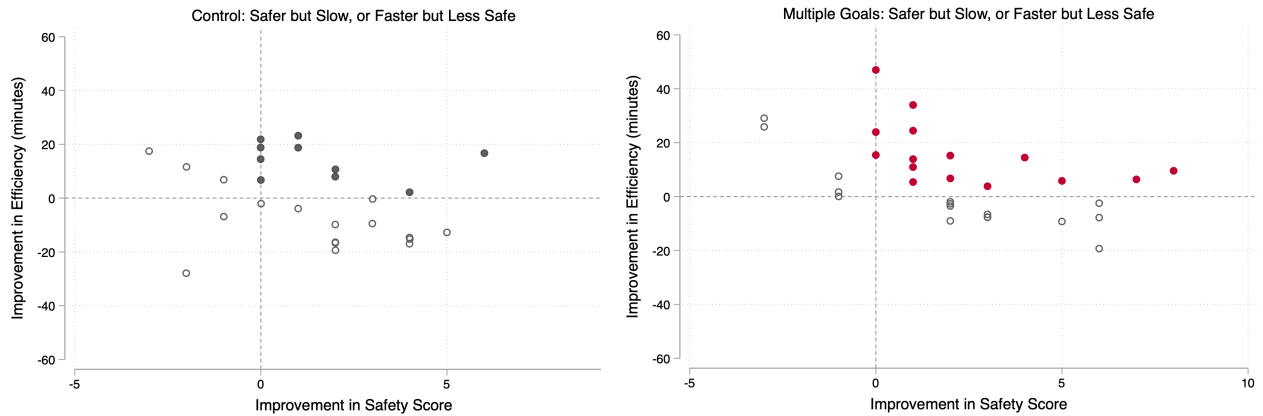


**FIGURE 5:** Improvements in efficiency and safety by baseline quadrant

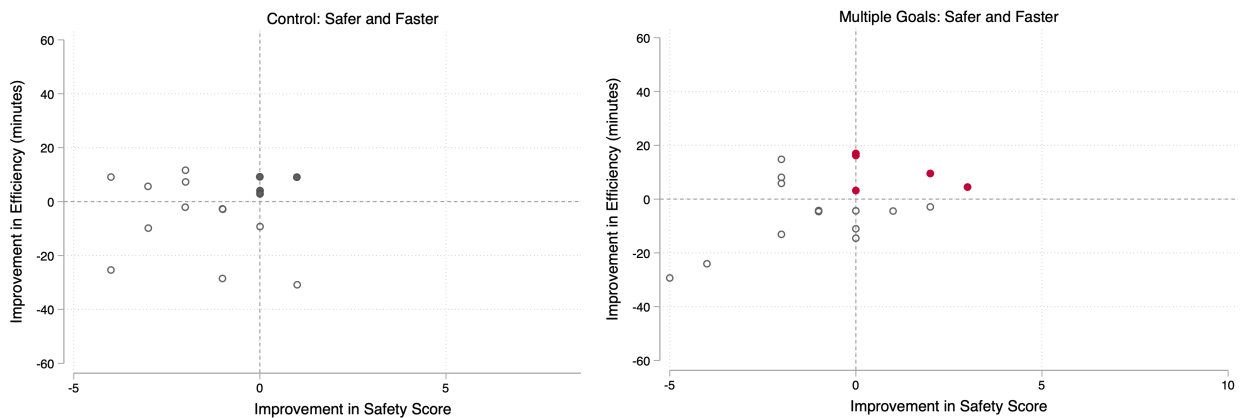
(a) Slower and Less Safe



(b) Safer but Slow, or Faster but Less Safe

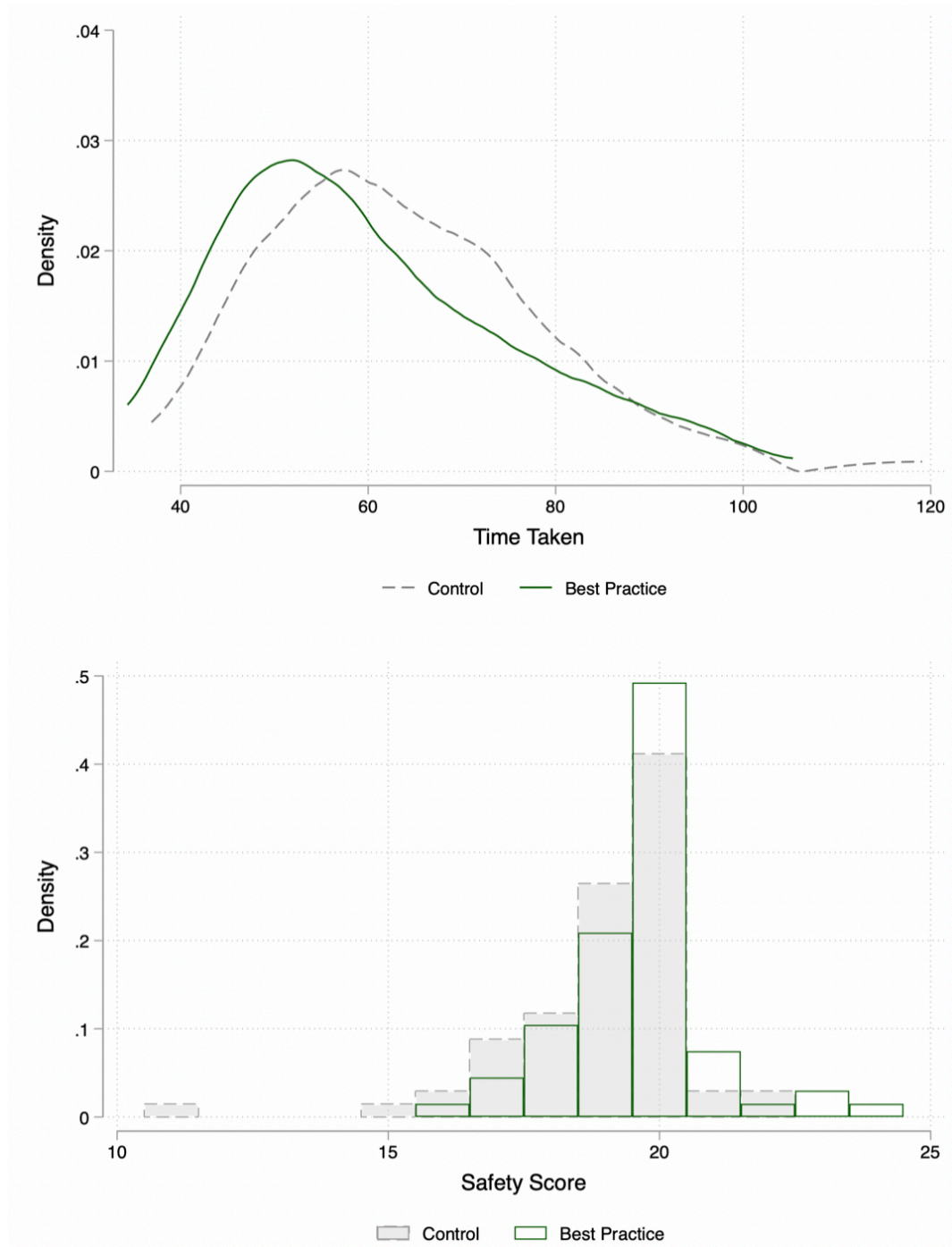


(c) Safer and Faster



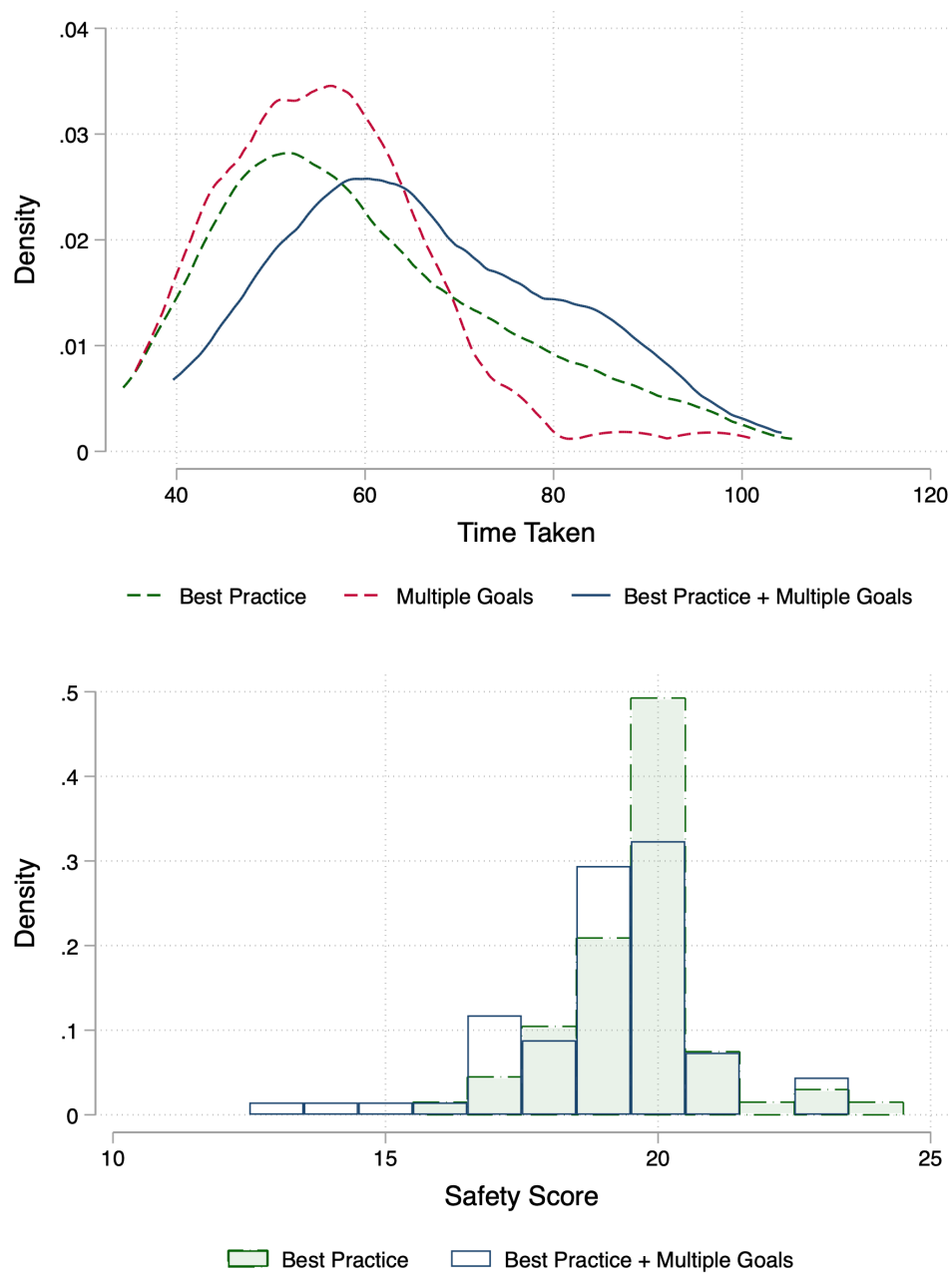
These figures show how linemen assigned to *Control* and *Multiple Goals* improve in efficiency and safety, segmented by their baseline performance compared to the median, as shown in quadrants in Figure 2(c): linemen who were slower and less safe at baseline, linemen who were safer but slow, or faster but less safe, and linemen who were both safer and faster. Filled-out circles indicate linemen pairs who improved their performance.

**FIGURE 6:** Average treatment effects of best practice observation



These figures plot the distribution of linemen pairs' times and safety scores in *Best Practice* compared to those in the *Control* group. Two-sample Kolmogorov-Smirnov tests for equality of distribution: Time: 0.22; Safety: 0.38.

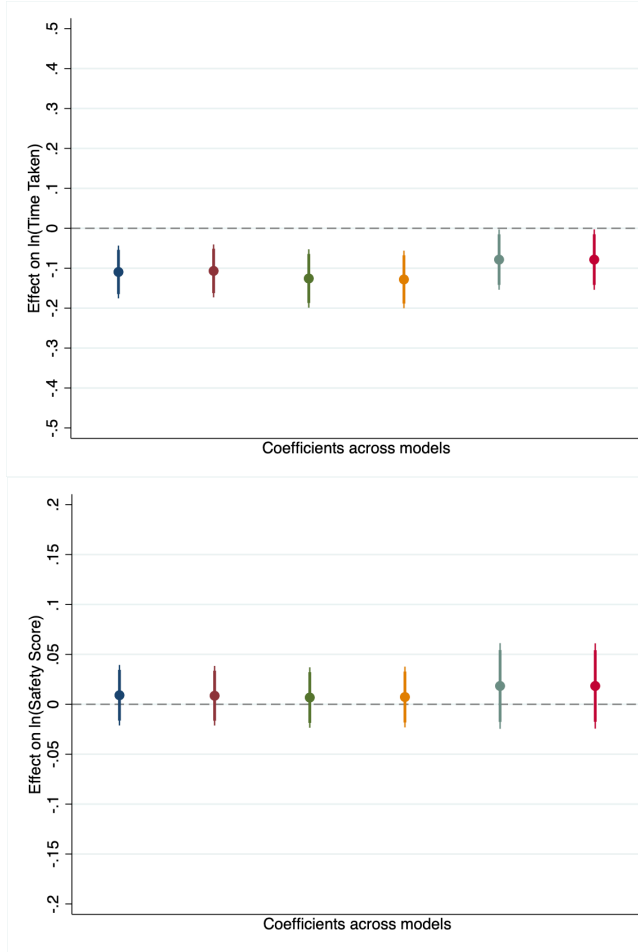
**FIGURE 7:** Average treatment effects of *Multiple Goals + Best Practice*



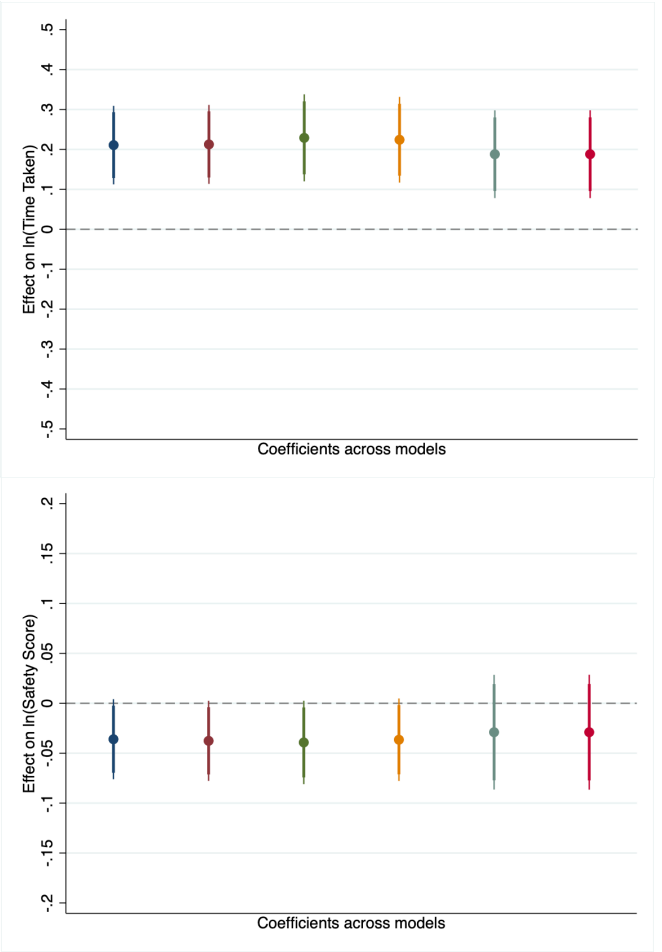
These figures plot the distribution of linemen pairs' times and safety scores in *Best Practice + Multiple Goals* compared to those in *Best Practice* (without Multiple Goals) in Round 2. Two-sample Kolmogorov-Smirnov tests for equality of distribution compared to *Best Practice*: Time: 0.01; Safety: 0.195.

**FIGURE 8:** Robustness of effects across econometric models

(a) The effect of *Multiple Goals*



(b) The effect of *Best Practice + Multiple Goals*



These figures plot treatment effect estimates on time taken (row 1) and safety (row 2) across different possible econometric models. The left column (a) plots the effects of *Multiple Goals*, while the right column (b) plots the effects of *Best Practice + Multiple Goals*. The first two coefficients in each figure come from the specification reported in the main text, with the first including the randomization stratum indicator and the second excluding it. The third and fourth coefficients estimate simple means of each outcome, with and without the randomization stratum indicator. The fifth and sixth models estimate coefficients from a difference-in-difference in model, with and without the randomization stratum indicator.